

Madhukosh

Bee Traceability Project — Onboarding Guide

How a jar of honey is traced from a Sundarbans mangrove hive to a consumer's hand, and how the Beekeeper, Cluster Collector, Admin, and Consumer each fit into that story.

The problem it solves

Honey is one of the most commonly adulterated food products in the world — sugar-syrup blending, mislabeled origin, and fake “raw forest honey” claims are widespread, and there is normally no way for a consumer to check whether the jar in their hand actually came from where the label says it did.

Madhukosh is a **traceability ledger**: it follows a batch of honey from a specific hive, in a specific forest location, harvested by a named person, all the way to the consumer scanning the bottle — and it automatically flags anything statistically suspicious along the way, such as a hive suddenly yielding four times its normal output.

The current build is themed around the **Sundarbans** mangrove forest delta, where traditional wild-honey collectors (“Moulis”) harvest from mangrove blooms such as Khalsi and Kewra, inside and around the Sundarban Tiger Reserve.

The four people in the system

Each persona maps to exactly one tab in the app. There is no login yet — see “What’s deliberately stubbed” at the end — so today anyone can act as any persona; the tabs exist to separate the workflows, not to restrict who can see them.

Persona	App tab	What they're trying to do
Cluster Collector	Cluster Collector	Field staff who onboard beekeepers and physically install / verify hive sensors in the forest.
Beekeeper	Harvest Logging	The Mouli who owns a hive and reports what they extracted from it.
Admin	Admin Dashboard	Back-office staff auditing the whole ledger for fraud and resolving flags.
Consumer	Batch Verification	The end buyer, scanning or typing a bottle's Batch ID to check its story.

The data spine: four tables, one chain of custody

Everything traces back through the same chain of custody. Every batch is really a join across all four tables — that is what makes the “provenance” story possible: a batch ID resolves all the way back to a named hive, a

named Mouli, a named collector, and a trust tier.



Every batch resolves back through this chain to a named hive, a named Mouli, and a named collector.

Table	What it represents
collectors	Regional field leads (e.g. Debashish Halder, “Cluster Lead — Sundarban Delta North”). The root of trust — nothing else references upward from here.
beekeepers	A Mouli, registered by a collector. Auto-assigned an ID like #BK-101.
hives	A physical hive box, tied to one beekeeper, provisioned by a collector. Auto-assigned an ID like HIVE-042, and carries a Trust Tier: Self-Attested (the collector’s word) or Lab-Verified (an actual sensor/lab check happened). New hives default to Self-Attested.
batches	One harvest event: this hive, this many kilograms, this date. Auto-assigned an ID like #MK-8921. This is the row a consumer ultimately looks up.

The end-to-end flow

1. Collector onboards a Mouli

Cluster Collector tab, card 1

The field collector registers a new beekeeper: name, phone, region. This is the first link in the chain — nothing downstream can exist without it, since hives require a `beekeeper_id`.

2. Collector provisions a hive

Cluster Collector tab, card 2

Picks the beekeeper from a dropdown and hits “Add a hive.” The system auto-generates the next HIVE-0xx ID server-side using a database sequence, so two collectors adding hives at the same moment never collide. New hives default to Self-Attested.

3. Collector verifies a hive

Cluster Collector tab, card 3

This is where trust is actually established. The collector picks an existing hive and asserts its tier: Lab-Verified or Self-Attested. This is not cosmetic — verifying a hive updates its trust tier live, and that tier is what shows up on **every** batch ever harvested from that hive, past and future, in the Admin ledger and the consumer-facing card. The hive's tier is authoritative, not something baked into each individual harvest.

4. Beekeeper logs a harvest

Harvest Logging tab

The Mouli picks their hive, enters kilograms and date, and submits. Two things happen automatically: the batch snapshots the hive's current trust tier at that moment, and a **fraud guard** checks the quantity — anything over 50kg from a single hive is automatically flagged as a “Yield Spike,” with status set to “Flagged for Anomaly Investigation.” This check runs twice: once in the app, and once as a database trigger, so it cannot be bypassed even by a direct API call. Submitting jumps straight to the Admin Dashboard so the new batch is immediately visible.

5. Admin audits the ledger

Admin Dashboard tab

The command center: every batch ever logged, newest first, color-coded — flagged rows get a red left-border and tint. Clicking any row opens the Audit Verification modal, the richest view in the app, showing the beekeeper's contact info, the collector's contact info, harvest/verification timestamps, and a simulated cryptographic hash standing in for a real ledger anchor. If a batch is flagged, the admin can hit “Resolve Anomaly Flag,” clearing the flag and writing a resolution note, as if a human inspector re-tested it.

6. Consumer verifies a bottle

Batch Verification tab

No login, no jargon — just a search box. Typing or scanning the Batch ID printed on the bottle (e.g. MK-8921) resolves the same join chain: which hive, which Mouli, which region, what purity tier. A “View full audit trail” link opens the exact same audit modal the admin uses — the same component, reused — so a curious consumer can see everything the admin sees, minus write access.

Why the flow is resilient to bad timing

Because collectors, beekeepers, and admins are all looking at the same shared database rather than separate local copies, actions in one tab immediately show up in another without a page reload. A lightweight publish/subscribe event bus in the app broadcasts events like “a hive was just created” or “a batch was just logged,” and any open tab that cares — for example, the Harvest Logging hive dropdown — refreshes itself automatically. Auto-generated IDs come from database sequences rather than “count the rows and add one,” so two people acting at the same instant can never collide.

What's deliberately stubbed right now

- **No authentication.** All four tabs are open navigation — anyone can act as any persona today. Writes are attributed to whichever collector happens to be first in the database, standing in for “the logged-in collector.” This was an explicit choice to prove out the data model and flow before adding login/roles.
- **No real PDF generation or QR scanning.** “Download Traceability Report” and the bottle-QR framing are UI placeholders — the batch lookup itself is fully real, just not the physical QR/PDF tooling around it.
- **The cryptographic hash is simulated,** not an actual blockchain anchor — it sells the “tamper-evident ledger” story visually, not a real cryptographic guarantee yet.